

Assignment 2

Part 1: Basic Slicing

1. Introduction to Slicing:

- a. Given a string `s = "Hello, World!"`, slice and print the following:
 - i. The entire string.
 - ii. The first 5 characters.
 - iii. The last 5 characters.
 - iv. The string excludes the first and last characters.
 - v. Every second character in the string.

2. Slicing with Negative Indices:

- a. Use negative indices to slice and print the following from the string `s`:
 1. The last 3 characters.
 2. All characters except the last 2.
 3. The string reversed.

Part 2: Advanced Slicing

1. Slicing Substrings:

- a. Given a string `sentence = "The quick brown fox jumps over the lazy dog"`, slice and print the following:
 - i. The word "quick".
 - ii. The word "lazy".
 - iii. The words "brown fox jumps"

2. Step Slicing:

- a. Use step slicing to print the following from the string `s`:
 - i. Every third character.
 - ii. Every third character starting from the second character.
 - iii. The string reversed, stepping by 2.

Part 3: Practical Application

1. Extracting Information:

- a. Given a string `data = "2024-09-01,Sunny,25°C"`, extract and print the following:
 - i. The date.
 - ii. The weather condition.
 - iii. The temperature

2. Reformatting Strings:

- i. Given a string `phone_number = "(123) 456-7890"`, reformat it to `'123-456-7890'`

Part 4: Questions based on string methods

1. Upper and Lower Case Conversion:

- i. Given a string `s = "Hello, World!"`, perform the following:
 - b. Convert the string to all uppercase letters.
 - c. Convert the string to all lowercase letters.
 - d. Convert the string to title case.
 - e. Capitalize the first letter of the string.

2. Finding Substrings:

Given a string `s = "The quick brown fox jumps over the lazy dog"`, perform the following:

- a. Find the position of the substring `"fox"`.
- b. Check if the substring `"cat"` is in the string.

3. Replacing Substrings:

Given a string `s = "The quick brown fox jumps over the lazy dog"`, perform the following:

- a. Replace `"fox"` with `"cat"`.
- b. Replace all spaces with underscores.
- c. Split the string into a list of words.
- d. Split the string by the letter `'o'`.

4. Joining Strings:

Given a list of words

`words = ["The", "quick", "brown", "fox"]`,

perform the following:

- a. Join the words with spaces.
- b. Join the words with hyphens